Thiyanayugi Mariraj

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LinkedIn GitHub Portfolio

Summary

AI Engineer and Robotics Specialist architecting the synapse between digital intelligence and physical action. Specializing in Autonomous Robotics, Generative AI, and Computer Vision with expertise in multi-agent systems, LLM frameworks (LangChain, CrewAI, AutoGen), and ROS-based robotic platforms.

Technical Skills

- **Programming:** Python, C++, MATLAB, Bash, JavaScript, SQL, Rust, Go, TypeScript
- **AI & ML:** PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, NumPy, Pandas, Jupyter
- **Generative AI:** LangChain, HuggingFace, CrewAI, AutoGen, n8n, RAG, Vector DB, Claude Code, REST API
- **Robotics:** ROS/ROS2, Isaac Sim, Gazebo, MoveIt, SLAM, LiDAR, PCL, CoppeliaSim, Sensor Fusion
- **Cloud & DevOps:** AWS, Google Cloud, Docker, Kubernetes, GitHub Actions, Grafana, Git, Linux
- **Computer Vision:** YOLOv8, CLIP, Object Detection, 3D Vision, Point Cloud Processing
- **Data Analysis:** Statistical Analysis, Time-Series, Data Visualization, Prometheus
- **Development:** FastAPI, Streamlit, Gradio, Web Development, REST APIs
- **Databases:** SQL, FAISS, ChromaDB, Graph Databases, Vector Search
- **Languages:** English (Fluent), German (B2 in Progress), Tamil (Native)

Education

Technical University Dortmund — **October 2022 – July 2026**
Master of Science in Automation and Robotics — *Dortmund, Germany*
- Specialization: AI-driven Systems, Computer Vision, Autonomous Robotics, 6G Collaborative Perception

PSG College of Technology — **July 2018 – May 2022**
Bachelor of Engineering in Robotics and Automation — *Coimbatore, India*
- Core Focus: Industrial Automation, Mobile Robotics, Control Systems, Embedded Systems

Professional Experience

ALMAWATECH GmbH — **February 2026 – Present**
AI Engineer – LLM & RAG Systems — *Babenhausen, Germany*
- Developing production AI systems for industrial water treatment using LLMs, RAG, and multi-agent frameworks.
- Building a knowledge management system for 25+ years of technical documents using vector DB and semantic search.
- Implementing LLM-based optimization tools for pilot studies and industrial process control.
- Tech: Python, LangChain, LangGraph, ChromaDB, FAISS, Claude API, OpenAI, FastAPI, Docker, GCP

Information Processing Laboratory, TU Dortmund — **June 2024 – May 2025**
Research Assistant – Machine Learning & Signal Processing — *Dortmund, Germany*
- Designed DL models for intelligent signal interpretation in complex electromagnetic systems achieving high accuracy.
- Applied reinforcement learning for adaptive optimization and real-time parameter tuning in dynamic environments.
- Improved signal processing precision in high-frequency industrial applications via supervised ML techniques.

Chair of Material Handling and Warehousing, TU Dortmund — **January 2025 – July 2025**
Master's Thesis – 6G Collaborative Robotic Perception — *Dortmund, Germany*
- Developed 6G-enabled collaborative perception for multi-robot systems using mmWave radar technology.
- Implemented Graph Neural Network architectures for real-time occupancy prediction in warehouse environments.
- Conducted comprehensive validation using dual robotic platforms with motion capture ground truth systems.

Pricol Limited — **November 2021 – June 2022**
Robotics Engineering Intern – Autonomous Systems — *Coimbatore, India*
- Designed and deployed Autonomous Mobile Robot for industrial logistics and inspection automation applications.
- Integrated LiDAR, IMU, and camera systems for robust SLAM-based navigation in dynamic industrial environments.
- Developed real-time path planning and obstacle avoidance algorithms using ROS framework.

Featured Projects

Autolearn AI Platform - Production Web Application

- Engineered production-ready web platform with Claude API-powered intelligent chatbot and AI-driven personalized email generation system for course enrollment.
- Deployed on Google Cloud Run with scalable architecture handling real-time user interactions.
- Technologies: JavaScript, Claude API, Google Cloud Run, REST APIs

AI Document Explainer - International Student Tool

- Built production-ready AI application helping international students understand complex official documents using GPT-4 natural language processing.
- Implemented Streamlit-based interactive interface with document upload, processing, and explanation features.
- Technologies: Python, GPT-4, Streamlit, Cloud Deployment

RoboVision-3D: Computer Vision for Indoor Robotics

- Developed comprehensive computer vision system integrating RGB-D cameras and LiDAR for 3D environment mapping and object detection.
- Implemented YOLOv8-based object detection with PCA-driven 3D localization processing over 19M point cloud data points.
- Engineered modular ROS-based architecture for real-time multi-sensor fusion and SLAM-based navigation.
- Technologies: Python, YOLOv8, ROS, OpenCV, Point Cloud Library

AutoGen Content Creation Pipeline - Multi-Agent AI

- Architected enterprise-grade multi-agent AI system automating end-to-end content creation workflows with specialized agents for research, writing, and editing.
- Implemented autonomous agent orchestration using Microsoft AutoGen framework with DeepSeek integration.
- Technologies: AutoGen, DeepSeek, Python, Multi-Agent Systems

Market Intelligence System - CrewAI Multi-Agent

- Developed CrewAI-powered multi-agent system with four specialized agents (Researcher, Analyst, Writer, Reviewer) for autonomous market intelligence gathering and analysis.
- Integrated Claude API and RAG techniques for enhanced reasoning and knowledge retrieval.
- Technologies: CrewAI, Claude, RAG, Python

GraphRAG - Knowledge Graph-Enhanced Retrieval System

- Architected production-ready hybrid retrieval system combining knowledge graphs with FAISS vector search for multi-hop reasoning and complex query resolution.
- Implemented automated entity extraction pipeline using spaCy NER with relationship mapping across documents.
- Achieved sub-second query response times processing 140+ entities with semantic relationship graphs.
- Technologies: Graph Databases, FAISS, Vector Search, Gradio, spaCy

Multi-Modal AI Framework for Robotic Task Automation

- Engineered multi-agent system integrating CLIP vision models, GPT-3.5 NLP, and RAG for intelligent robotic task planning and execution.
- Implemented cross-modal understanding enabling robots to process visual and textual instructions simultaneously.
- Technologies: LangChain, CLIP, GPT-3.5, RAG, Computer Vision

Edge Detection 3D Vision System

- Built 3D vision system with real-time image processing and modular edge detection algorithms using ROS framework.
- Integrated OpenCV and RViz for visualization and processing pipeline optimization.
- Technologies: ROS, OpenCV, RViz, 3D Vision

Certifications

- **AWS Training:** Generative AI Learning Plan for Developers, Amazon SageMaker JumpStart Foundations, No-code Machine Learning and Generative AI on AWS, Building Language Models on AWS
- **Hugging Face:** AI Agents with HuggingFace - Fundamentals of AI Agents
- **DeepLearning.AI:** ChatGPT Prompt Engineering for Developers
- **LinkedIn Learning:** What Is Generative AI?, Generative AI: The Evolution of Thoughtful Online Search

Publications

- Mariraj, T., et al. (2021). **RFID based Human Following Load Carrier**. NVEO - Natural Volatiles & Essential Oils Journal, 8(5). [Link]
- Mariraj, T., et al. (2021). **Development Of Robotic Arm On An Omnidirectional Base**. NVEO - Natural Volatiles & Essential Oils Journal, 8(4). [Link]